Reviewer Response

Handeling editor

Dear Authors,

I have now received 2 reviews of your manuscript. As you can see, both referees found the manuscript to be interesting and well executed but also raised a number of concerns. These involve a need for more detail on the evidence for some claims (e.g. the effect of diversification on beta diversity), improved explanation of methods (e.g. random effects), consideration of possible confounders, and suggestions around statistical analyses/results reporting to improve clarity.

> Response
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> We thank the editor and reviewers for their work in providing all feedback.

All of these should be, in my view, addressable in a revised version, which I look forward to seeing.

Please see below how we address all points.

I have one additional point - I appreciate the custom package to carry out the analysis, and having it on GitHub is fine for peer review, but please note that GitHub does not meet GEB's policies on stable access for Code and Data (see: https://onlinelibrary.wiley.com/page/journal/14668238/homepage/forauthors.html), so you will need to have an alternative should the paper be accepted.

Understood and not to worry, our plan is to release the R package via zenodo

The authors collected arthropod abundance and body size data across regions of different geological age over the Hawaiian Archipelago and examined the observed species abundance distribution and the individual power distribution (the metabolic rate across individuals) against the expected distribution under the Maximum Entropy Theory of Ecology. They found that deviations from the METE expectation strengthens in regions of middle age, which have increased beta-diversity. The authors attributed the deviation from the METE expectation,

which is made under the assumption of equilibrium, to adaptive radiation and the introduction of non-native species.

I much appreciate the authors effort to not only check the observed macroecological patterns against the METE expectation, but also associate the deviation from the METE expectation to environmental and ecological factors. However, I have some minor concerns regarding the presented results:

1. The authors' main quantification of the deviation from METE was the z^2 value calculated from the observed log likelihood of data and the log likelihood of data simulated under METE. They associated the deviation from METE to environmental and ecological factors by applying linear regressions on log-transformed z^2. The authors also quantified the deviation using difference in dominance or rarity. I wonder if the authors may consider the adjusted R^2 as described by Shoemaker et al (https://doi.org/10.1038/s41559-017-0107) where the predicted rank-abundance distribution is compared against the observed one as an alternative and potentially more intuitive, straightforward quantification for illustration and statistical analysis.

2. The authors checked the deviation against METE, but appeared to not show any statistics on whether METE can sufficiently describe the observed pattern? If I understand the z^2 correctly, 95% of the time a simulated data under METE will have a z^2 <3.84, which the z^2 from many sites fell below. The significance of deviation from METE would be very helpful to establish the basis for following association with environmental and ecological factors. It would also be interesting to see if the deviation of SAD and IPD from METE within each site is correlated.

3. For Figure 3, the authors showed two exemplar cases to show deviation. I wonder if the authors may use the rank-paired scatter plot as used by Shoemaker et al or Gao et al (https://doi.org/10.1038/s41467-025-59253-9) to show the difference across all cases (e.g., faceted by sites)? This would be a good addition to show the deviation from METE expectations.